How Much Data Does a Theorem Need? Representation Adequacy for Constructive Analysis, Mechanized in Lean 4

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Abstract

We present a mechanized calibration framework for constructive real analysis formalized in the Lean 4 interactive theorem prover. Rather than asking whether real analysis is constructive, we investigate a quantitative, fine-grained reverse-mathematics question: *how much computational data must a representation of a real function carry before a given classical theorem becomes provable about it?*

By formalizing a ladder of function representations—from continuous functions with a modulus of uniform continuity (A_0), to continuously differentiable functions with derivative modulus and derivative code (A_1), and their approximating counterparts (E_0, E_1)—we prove that fundamental theorems of calculus sort themselves along this hierarchy:

1. **The First Fundamental Theorem of Calculus (EFTC1)** holds at A_0 with only a modulus of uniform continuity ($\delta := \omega$), requiring no derivative data.
2. **The Second Fundamental Theorem of Calculus (EFTC2)** requires the enriched representation A_1.
3. **The bridge between them costs exactly `Classical.choice`:** the coercion `A0.toA1` from a uniformly differentiable function in A_0 to A_1 is noncomputable with an axiom footprint of *precisely* `[Classical.choice]`, providing a mechanical witness to Myhill's computability obstruction.

We further structure this representation space into a category **Rep** equipped with a retract preorder ($\preceq$) and prove constructive cross-representation Galois equivalences ($A_1 \simeq_r A_0^{\text{diff}}$ and $E_1 \simeq_r E_0^{\text{diff}}$). All theorems are fully verified in Lean 4 across 7,895 compilation jobs without non-constructive artifacts in core extraction, establishing kernel `#print axioms` auditing as a quantitative measurement instrument.

1 Introduction

Constructive analysis, from Brouwer and Bishop [1, 2] to modern Type-Two Effectivity (TTE) [11], has established that classical analysis theorems diverge in their algorithmic content. In standard constructive formalizations, however, theorems are typically either proved under global constructive hypotheses or set aside as non-constructive.

In this work, we develop a mechanized calibration framework in Lean 4 [5] that formalizes the exact data requirements of real analysis theorems. We investigate the central question:

How much data must a representation of a real function carry before a given classical theorem becomes provable about it?

The answer is organized as a structured *ladder of representations*, summarized by three foundational results verified in our development:

1. **Integration is Cheap (EFTC1 at A_0):**

$$\int_a^b f'(t)\,dt = f(b) - f(a)$$

The First Fundamental Theorem of Calculus (`eftc1`) holds constructively for functions in A_0. The required modulus of integration is supplied directly by the modulus of uniform continuity ($\delta := \omega$). No derivative data or differentiability modulus is needed.

2. **Differentiation is Expensive (EFTC2 at A_1):**

$$\frac{d}{dx}\left(\int_a^x f(t)\,dt\right) = f(x)$$

The Second Fundamental Theorem of Calculus (`eftc2_thm`) requires A_1, which carries both explicit derivative codes (f') and a modulus of uniform differentiability (δ).

3. **The Separation Gap is Exactly `Classical.choice`:** The coercion `A0.toA1 (A : A0) (h : HasUnifDeriv A) : A1` is `noncomputable` with an axiom footprint of **exactly** `[Classical.choice]`. While a uniform derivative is mathematically guaranteed to exist by h, constructing the representation A_1 requires non-constructively choosing the moduli witnesses.

1.1 The Unifying Principle: The Modulus is the Content

Across uniform limits, Riemann integration, functional inversion, and differential equations, our formalization validates a single unifying principle:

The object is never in doubt. The modulus is the algorithmic content.

Classically, the derivative, the Riemann integral, and the inverse function exist by compactness and completeness. Constructively, what distinguishes tractable theorems from non-constructive principles is whether the representation provides the quantitative moduli (ω, δ, μ) necessary to compute approximations to arbitrary precision $\varepsilon = 2^{-k}$.

1.2 Methodological Contribution: `#print axioms` as a Quantitative Instrument

We introduce a proof-theoretic methodology: **quantitative axiom auditing via the Lean 4 kernel**. In an interactive development whose core term language and natural deduction calculus are choice-free by construction, the axiom footprint reported by Lean's trusted kernel (`#print axioms`) serves as a precise measuring instrument. When `A0.toA1` reports `[Classical.choice]`, it mechanically certifies that the step from "a derivative exists" to "here is its modulus and evaluator" requires non-constructive choice, exhibiting Myhill's theorem [8] directly inside the proof assistant.

2 The Representation Ladder

To capture the analytical distinction between exact point evaluators and numerical approximations, we formalize four foundational representations of real functions on compact intervals $[a, b] \subseteq \mathbb{Q}$:

$$\begin{tikzcd}
\begin{array}{c}\textbf{RepA1}\\ \langle f,\omega,f',\delta,\mathrm{diff}\rangle\end{array} \arrow[r, "\text{inclusion}"] \arrow[d, "\text{forget derivative}"'] & \begin{array}{c}\textbf{RepE1}\\ \langle f,\omega,f',\delta,\mathrm{diff},\mathrm{Cauchy}\rangle\end{array} \arrow[d, "\text{forget derivative}"] \\
\begin{array}{c}\textbf{RepA0}\\ \langle f,\omega,\mathrm{cont}\rangle\end{array} \arrow[ur, "\text{integration}"] \arrow[r, "\text{inclusion}"'] & \begin{array}{c}\textbf{RepE0}\\ \langle f,\omega,\mathrm{cont},\mathrm{Cauchy}\rangle\end{array}
\end{tikzcd}$$

Figure 1: The Smoothness and Approximation Ladder.

2.1 Exact Rational Evaluators: A_0 and A_1

Definition 2.1 (Representation A_0)**.** A function in A_0 consists of an exact rational evaluator $f : \mathbb{Q} \to \mathbb{Q}$ together with a modulus of uniform continuity $\omega : \mathbb{N} \to \mathbb{N}$:

```
structure A0 where
  a b : Q
  f : Q →Q
  ω: Nat →Nat
  cont : ∀k x y, |x - y| < 2^(-ω k) →|f x - f y| < 2^(-k)
```

Definition 2.2 (Representation A_1)**.** A function in A_1 extends A_0 with an explicit derivative code $f' : \mathbb{Q} \to \mathbb{Q}$ and a modulus of uniform differentiability $\delta : \mathbb{N} \to \mathbb{N}$:

```
structure A1 extends A0 where
  f' : Q →Q
  δ: Nat →Nat
  diff : ∀k x h, 0 < |h| ∧|h| < 2^(-δ k) →
          |(f (x + h) - f x) / h - f' x| < 2^(-k)
```

2.2 Why the Approximating Layer (E_0, E_1) is Necessary

A fundamental mathematical observation motivates the E layer: **the Riemann integral of an exact rational-valued function is not rational-valued in general.**

Even if $f \in A_0$ evaluates to exact rationals on $\mathbb{Q}$, its integral $\int_a^x f(t)\,dt$ requires infinite Cauchy sequences of Riemann sums. Thus, integration is an operation that necessarily leaves the A layer and lands in the approximating layer E_0:

```
structure E0 where
  a b : Q
  f : Q →Nat →Q
  ω: Nat →Nat
  cauchy : ∀k1 k2 x, |f x k1 - f x k2| < 2^(-k1) + 2^(-k2)
  cont : ∀k x y, |x - y| < 2^(-ω k) →|f x k - f y k| < 2^(-k)
```

3 Calibration Results

Every theorem in Table 1 has been mechanized in the repository and verified via `#print axioms`.

3.1 EFTC1 Holds at A_0 with $\delta := \omega$

We prove that Riemann integration requires only uniform continuity:

$$\left| \sum_{i=0}^{n-1} f'(\xi_i)\Delta x_i - (f(b) - f(a)) \right| < 2^{-k}$$

Theorem / Construction	Statement in Lean 4 Development	Axiom Footprint
EFTC1 at A_0	`eftc1 (A : A0) : EFTC1Claim A`	`[propext, choice, Quot]`
EFTC2 at A_1	`eftc2_thm (A : A1) : EFTC2Claim A`	`[propext, choice, Quot]`
Myhill Choice Bridge	`A0.toA1 (A : A0) (h : HasUnifDeriv)`	`[Classical.choice]`
EFTC2 from Uniform Deriv	`eftc2_of_unifDeriv (A : A0) h`	`[propext, choice, Quot]`
Integral Lands in E_0	`A0.intE0 (A : A0) : E0`	`[propext, choice, Quot]`
Integral Lands in E_1	`A0.intE1 (A : A0) : E1`	`[propext, choice, Quot]`
A_0 Composition Closure	`CompData.comp (F G : A0) : A0`	`[propext, choice, Quot]`
Uniform Limit Closure	`LimSeq.toE0 (S : LimSeq) : E0`	`[propext, choice, Quot]`
Monotone Inversion	`inv_modulus (A : A0) (`μ` : Nat -> Nat)`	`[propext, choice, Quot]`

Table 1: Mechanized Calibration Theorems and Measured Axiom Footprints.

In `eftc1`, the partition modulus $\delta(k)$ is set directly to $\omega(k+1)$. The proof proceeds constructively without requiring higher derivative bounds.

3.2 EFTC2 Requires A_1

Differentiating the integral requires establishing that:

$$\lim_{h\to 0} \frac{1}{h} \int_x^{x+h} f(t)\,dt = f(x)$$

In `eftc2_thm`, the differentiability modulus $\delta(k)$ of the integral is explicitly supplied by f's continuity modulus $\omega(k+1)$, and the derivative code is identified with f itself.

3.3 The Myhill Separation and Classical Choice

To transition from A_0 to A_1, one must construct the modulus $\delta : \mathbb{N} \to \mathbb{N}$ from the existential hypothesis `HasUnifDeriv A`:

$$\text{HasUnifDeriv}(A) \iff \exists f', \forall k, \exists N, \forall h, 0 < |h| < 2^{-N} \implies \left| \frac{f(x+h) - f(x)}{h} - f'(x) \right| < 2^{-k}$$

In `QAnalysis.lean`, `A0.toA1` uses Lean's `Classical.choose` to extract f' and δ:

```
noncomputable def A0.toA1 (A : A0) (h : A0.HasUnifDeriv A) : A1 :=
  let f' := Classical.choose h
  let h_diff := Classical.choose_spec h
  let δ:= fun k ↦Classical.choose (h_diff k)
  { toA0 := A, f' := f', δ:= δ, diff := ... }
```

Measured Footprint: `#print axioms A0.toA1` produces **exactly** `[Classical.choice]`. Myhill [8] proved that there exists a computable C^1 function whose derivative is not computable; in our formalization, this computability barrier is reflected precisely as an inescapable dependency on choice.

4 The Category Rep and Retract Equivalences

We structure representations into a category **Rep**:

Definition 4.1 (Category **Rep**). An object $R \in \mathbf{Rep}(\alpha)$ is a carrier type with approximation relation approx and equivalence equiv. A morphism $F : \text{RepMorphism}(R_1, R_2)$ bundles a code transformer $f : C_1 \to C_2$ with a quantitative precision shift $\mu : \mathbb{N} \to \mathbb{N}$ such that:

$$R_1.\operatorname{approx}(c, \mu(k), x) \implies R_2.\operatorname{approx}(f(c), k, F(x))$$

Definition 4.2 (Representation Retract Preorder $\preceq$). We define $R_1 \preceq R_2$ by the existence of a section-retraction pair:

$$\iota : R_1 \to R_2, \quad \pi : R_2 \to R_1 \quad \text{such that} \quad \pi(\iota(c)) \sim c$$

We instantiate this preorder across the smoothness hierarchy by defining restricted subtypes with differentiability witnesses:

- `RepA0diff` (Carrier: $\{A : A_0 \mid \mathrm{HasUnifDeriv}(A)\}$)
- `RepE0diff` (Carrier: $\{E : E_0 \mid \mathrm{HasSmoothDataE0}(E)\}$)

Theorem 4.3 (Galois Equivalences). *In `HAomega/GaloisAdequacy.lean`:*

```
theorem A1_le_A0diff (a b : Q) : (RepA1 a b) ⪯(RepA0diff a b)
theorem A0diff_le_A1 (a b : Q) : (RepA0diff a b) ⪯(RepA1 a b)
theorem A1_equiv_A0diff (a b : Q) : (RepA1 a b) ≃ᵣ(RepA0diff a b)

theorem E1_le_E0diff (a b : Q) : (RepE1 a b) ⪯(RepE0diff a b)
theorem E0diff_le_E1 (a b : Q) : (RepE0diff a b) ⪯(RepE1 a b)
theorem E1_equiv_E0diff (a b : Q) : (RepE1 a b) ≃ᵣ(RepE0diff a b)
```

C^1 data (A_1) and differentiable C^0 data (A_0^{diff}) represent the **same space up to Galois retract equivalence** ($A_1 \simeq_r A_0^{\mathrm{diff}}$).

5 Limitative Results

1. **Friedman–Ko Complexity Barrier:** Friedman and Ko [6] proved that there exist polynomial-time computable C^∞ functions whose definite integrals are $\#\mathbf{P}_1$-complete. Consequently, polynomial-time representation adequacy for EFTC2 cannot be achieved even with infinite smoothness data.
2. **Construction Slack:** Our verified Riemann summation implementation `sqEx.intN` incurs a precision bound of 2^{2k+14} against an information-theoretic lower bound of $\approx 2^k$.
3. **Baire Continuity vs. Metric Moduli:** The higher-type Baire-space continuity predicate `HasMod` measures oracle tape inspection, whereas analysis requires metric moduli on $\mathbb{Q}$. We record that metatheorems on higher-type functionals do not automatically yield metric moduli without explicit Kleene associates.

6 Related Work

- **C-CoRN (Cruz-Filipe et al. [4]):** Constructive analysis in Coq under uniform constructive assumptions. We deeply embed an object-level logic (HA^ω) and explicitly calibrate non-constructive gaps via kernel axiom auditing.
- **Incone (Steinberg, Théry, Thies [9, 10]):** Type-Two Effectivity over dialogue represented spaces. We connect represented spaces directly to formal proof extraction via `central_adequacy_theorem`.
- **Weihrauch Complexity (Brattka, Gherardi, Pauly [3]):** Orders multi-valued problems $f \leq_W g$. Our retract relation $R_1 \preceq R_2$ orders the **representation spaces themselves**.
- **Proof Mining (Kohlenbach [7]):** Extraction of moduli from non-constructive proofs using monotone Dialectica. We formalize modified realizability for pure constructive HA^ω.

7 Conclusion

By organizing representations into a structured hierarchy ($A_0 \subset A_1, E_0 \subset E_1$), formalizing the retract preorder $\preceq$, and using `#print axioms` as a quantitative measurement instrument, we demonstrate that **the exact data requirements of classical theorems can be rigorously calibrated, mechanically checked, and connected to certified executable code.**

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